Assymex Monterrey S.A. de C.V.

QUALITY CONTROL REPORT (10D)

QCR	18799	*Date:*	24-09-2024
AssyMex Contact / Title:	Leticia Castillo	*Customer Contact / Title:*	Martin Sosa
Contact Phone:		*Contact Phone:*	8128607691
Email:	leticia.castillo@avocarbon.com	*Contact Email:*	martin.sosa@tricoproducts.com

ONLY FILL THIS SECTION FOR INTERNAL SITUATIONS DETECTED AT ASSYMEX PLANT

- ☐ **INTERNAL PPM**
- ☐ **EXTERNAL AUDIT BY CUSTOMER**
- ☐ **EXTERNAL AUDIT BY IATF**
- ☐ **INTERNAL AUDIT BY IATF**
- ☐ **INTERNAL METRIC:** _____
- ☐ **OTHER:** _____

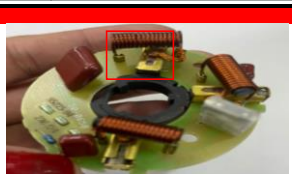
ONLY FILL THIS SECTION FOR QUALITY PROBLEMS DETECTED WITH CUSTOMER

- ☐ **Level 1 - Customer Return** - Immediate containment actions are required for all In House and Incoming stock within 24 hrs. Product Return Approval (RMA) may be necessary.
  Description of Root Cause & Corrective Action Plan are mandatory within 5 days.
- ☐ **Level 2 - Quality Rejection** - Immediate containment actions are required for all In House and Incoming stock within 24 hrs. Product Return Approval (RMA) may be necessary.
  Description of Root Cause & Corrective Action Plan are mandatory within 5 days.
- ☐ **Level 3 - Quality Concern** - Description of Root Cause & Corrective Action Plan are mandatory within 5 days.
  All Incoming stock 100% certified.
- ☐ **Level 4 - Notification** - All Incoming stock 100% certified.

WHAT IS THE PROBLEM ? > Use 4W + 2H to know what is the Current Situation ? Customer View and AssyMex View

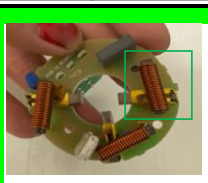
- What happened? *Description of the Problem*	**Brushcard without ferrite on inductor**	**- Why is it a problem?**	**Low perfomance of motor (low inductance)**

BAD PART



Lack of ferrite.

GOOD PART



With ferrite on inductor

- When it happened? *Date: dd-mm-yyyy*	**24-09-2024**	**- Where detected?** *Incoming, Production Line, Customer, Warranty, etc.*	**In production line Trico**
- How detected?	**Visualy**	**- How many bad parts?**	**1**

Additional Information for Customer

Customer Part Number:	**95015-94**	*AssyMex Part Number:*	**33-0152**
Customer Lot Number:	**23/08/2024**	*AssyMex Lot Number:*	**´23624**
Customer Inventory:	**1,344**	*AssyMex Inventory:*	**1300**
In Transit Inventory:	**0**	*Logistic Platform Inventory:*	**0**

Material Disposition

- ☐ Return to Supplier. RMA #
- ☐ Sort by Supplier.
- ☐ Sort / Rework by External Company WO #
- ☐ Scrap. Supplier's expense.
- ☐ Use as received
- ☐ Sort / Rework by AssyMex WO #

CORRECTIVE ACTION REPORT - 10D REPORT

1.- Problem Solving Team Members

Leader:	Leticia Castillo	**Title:**	Quality Enngenier	**Phone:**	8117945185
Email:	leticia.castillo@avocarbon.com	**Fax:**		**Mobile:**	

Group: Sughey Rangel, Matilde Serrano, Marco Estrada, Ramses Rendon, Alicio Flores

2.- Describe the Problem

Inductor without ferrite

- Was part reworked by supplier ?	NO	**- When was it manufactured ?**	24/09/2024
- Are we capturing the defect when reinjecting product in normal process ? If YES, explain how ?	**NO**	**- Did a similar problem happen previously at AssyMex or internally ? If YES, when ?**	**NO**

3.- Immediate Countermeasures / Problem Containment ? (24 hours)

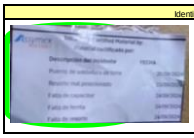
IMMEDIATE COUNTERMEASURES IN SUPPLIER PERIMETER	RESPONSIBLE	DATE	WHERE?
Inspection visual 100%	Daniel Morales	24/09/2024	Trico Plant

IMMEDIATE COUNTERMEASURES IN AssyMex PERIMETER	RESPONSIBLE	DATE	WHERE?
Inspection visual 100%	Leticia Castillo	24/09/2024	Assymex

Sorting Activity >> by Supplier

	Total Sorted	Total Bad Parts		PPM	Sorting Date	Responsible
- Suspect Inventory in Customer plant :	1344	4	=	2976.190476	24/09/2024	Daniel Morales
- Suspect Inventory in AssyMex plant :	1300	0	=	'----'	24/09/2024	Leticia Castillo
- Suspect Inventory in Transit :	NA		=	'----'		
- Suspect Inventory in Logistic platform :	NA		=	'----'		
- Suspect Inventory in Supplier plant :	0	0	=	'----'		

Identification of OK Parts

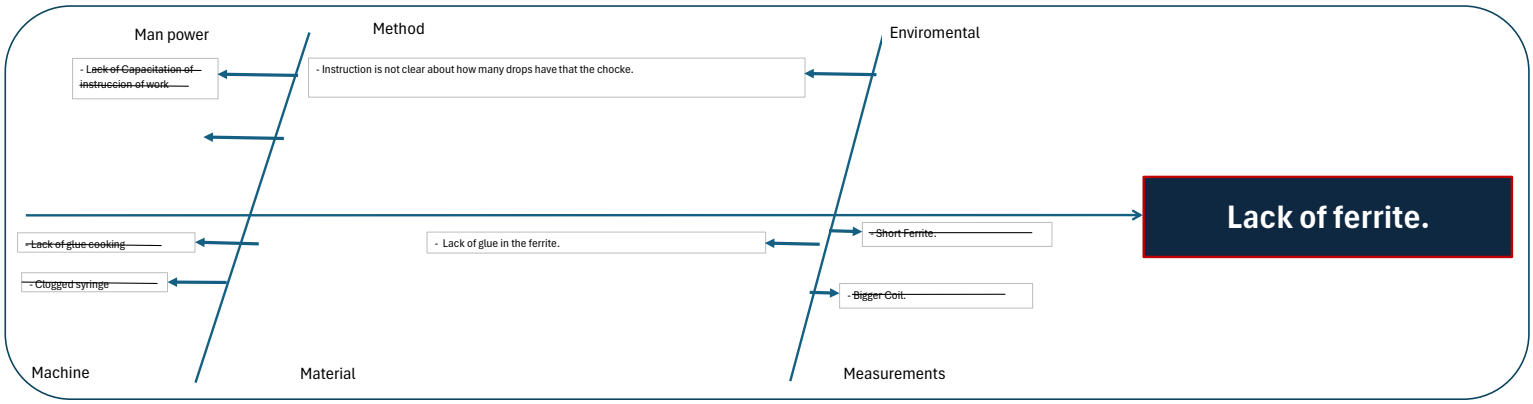


Certification label per box

	From:	To:
- Production dates of sorted parts:	23-08-2024	24-09-2024
- Production dates of bad parts:	Oldest part: 23-08-2024	Most recent part: 23-08-2024

First Certified Good Shipment	Lot Number:
	269
	Date: 24-09-2024

4,1.-	Analysis and Root Cause - Ocurrence



5 x Why Analysis Form - Ocurrences
Why did it happen?

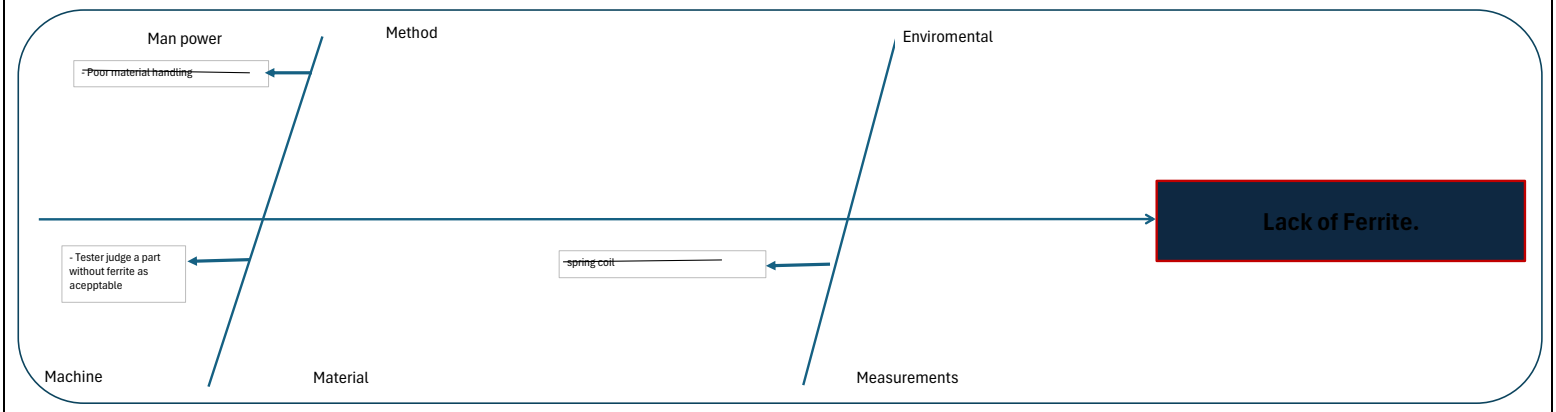
Cause: Instruction is not clear about how many drops have that the chocke.	
1st Why?	Question: Why lack ferrite in chocke?
	Answer: because the process was not executed correctly
2nd Why?	Question: Why the process was not executed correctly
	Answer: Because it does not mention the step by step of how many drops of glue the ferrite should have.
3rd Why?	Question: Why it does not mention the step by step of how many drops of glue the ferrite should have.?
	Answer: Because it changed the assembly process.
4th Why?	Question: Why Because it changed the assembly process?
	Answer: Because we sought to improve the efficiency of the process.
5th Why?	Question:
	Answer:

Cause: Lack of glue in the ferrite.	
1st Why?	Question: Why lack ferrite in chocke?
	Answer:Because it lacks glue.
2nd Why?	Question: Why it lacks glue?
	Answer: Because the drop of glue was not the right one.
3rd Why?	Question: Why the drop of glue was not the right one.?
	Answer: Because the glue did not have the correct temperature.
4th Why?	Question: Why the glue did not have the correct temperature?
	Answer: Because there was no preheating and temperature control of the glue
5th Why?	Question: Why there was no preheating and temperature control of the glue
	Answer: Work instructions did not consider preheating and temperature control of the glue.

Cause:	
1st Why?	Question:
	Answer:
2nd Why?	Question:
	Answer:
3rd Why?	Question:
	Answer:
4th Why?	Question:
	Answer:
5th Why?	Question:
	Answer:

Cause:	
1st Why?	Question:
	Answer:
2nd Why?	Question:
	Answer:
3rd Why?	Question:
	Answer:
4th Why?	Question:
	Answer:
5th Why?	Question:
	Answer:

4,2 .-	Analysis and Root Cause - Non Detection



5 x Why Analysis Form - Non-Detection
Why did the non-conformity escape?

Cause: Tester judge a part without ferrite as acepptable	
1st Why?	Question: Why tester judge a part without ferrite as acepptable?
	Answer:Because it doesn't do the inductance test.
2nd Why?	Question: why it doesn't do the inductance test?
	Answer: Because the tester does not have inductance equipment.
3rd Why?	Question: Why the tester does not have inductance equipment.?
	Answer: Because it was not considered in the test of tester.
4th Why?	Question:
	Answer:
5th Why?	Question:
	Answer:

Cause:	
1st Why?	Question:
	Answer:
2nd Why?	Question:
	Answer:
3rd Why?	Question:
	Answer:
4th Why?	Question:
	Answer:
5th Why?	Question:
	Answer:

Cause:	
1st Why?	Question:
	Answer:
2nd Why?	Question:
	Answer:
3rd Why?	Question:
	Answer:
4th Why?	Question:
	Answer:
5th Why?	Question:
	Answer:

Cause:	
1st Why?	Question:
	Answer:
2nd Why?	Question:
	Answer:
3rd Why?	Question:
	Answer:
4th Why?	Question:
	Answer:
5th Why?	Question:
	Answer:

5.- Corrective Action

To Eliminate Occurrence of the Problem:

Corrective Action	Expected result efficiency	Who	Due Date	Done Date
Preheat the glue before starting the operation.	Parts without loose ferrites.	Matilde Serrano	24-sep-24	24-sep-24
Update instruction with the number of drops of glue you should have.	Parts without loose ferrites.	Leticia Castillo	24-sep-24	27-sep-24
Add the preheating of the glue to the instructions.	Parts without loose ferrites.	Leticia Castillo	24-sep-24	27-sep-24

To Improve Detection of the Problem:

Corrective Action	Expected result efficiency	Who	Due Date	Done Date
Add inductance test on capacitance tester	Parts without loose ferrites.	Alicio Flores	02-dic-24	

6,1.- Implement and Verify Corrective Action - Occurences

Checks	CONFIRMATION CHECKS	CHECK QUALITY PER SHIFT 1st	2nd	3rd
Confirmation of influent Factors	1- HAVE PARAMETERS BEEN CORRECTED IN SAFETY / WORKING INSTRUCTIONS?	Y		
	2- DO WORKING INSTRUCTIONS MATCH WITH QUALITY CONTROL PLAN?	Y		
	3- IS ALL DOCUMENTATION COHERENT?	Y		
	4- ARE THE INSTRUCTIONS CLEAR?	Y		
	5- ARE CRITICAL POINTS CLEARLY DESCRIBED?	Y		
Physical Check of Operation	6- DOES ACTUAL PROCESS MATCH WITH THE OPERATING INSTRUCTIONS?	Y		
	7- DOES ACTUAL PROCESS MATCH WITH EXPECTED CORRECTIVE ACTION?	Y		
	8- ARE OPERATORS AWARE OF THE DEFECT?	Y		
	9- ARE OPERATORS AWARE OF THE EXPECTED CORRECTIVE ACTION?	Y		
	10- HAS THE SOLUTION BEEN PROPERLY VALIDATED ON THE FLOOR?	Y		
Visual Aid	11- IS THERE A VISUAL AID?	Y		
	12- IS IT POSTED ON A VISIBLE SPOT?	Y		
	13- IS THE CONTENT CLEAR?	Y		
Equipment	14- IS IT COHERENT WITH THE CONTENT OF THE CORRECTIVE ACTION?	Y		
	15- HAS IT BEEN MODIFIED? IS PREVENTIVE MAINTENANCE RECORD MODIFIED?	N/A		
	16- IS SET UP INSTRUCTION UPDATED? MATCHES WITH CONTROL PLAN?	N/A		

AUDITOR'S NAME		DATE OF THE AUDITS	27-09-2024					
Comments		DECISION	Y N	Y	Y N		Y N	

6,1.- Implement and Verify Corrective Action - Non - Detection

Checks	CONFIRMATION CHECKS	CHECK QUALITY PER SHIFT 1st	2nd	3rd
Confirmation of Influential Factors	1- HAVE PARAMETERS BEEN CORRECTED IN SAFETY / WORKING INSTRUCTIONS?	N		
	2- DO WORKING INSTRUCTIONS MATCH WITH QUALITY CONTROL PLAN?	N		
	3- IS ALL DOCUMENTATION COHERENT?	N		
	4- ARE THE INSTRUCTIONS CLEAR?	Y		
	5- ARE CRITICAL POINTS CLEARLY DESCRIBED?	Y		
Physical Check of Operation	6- DOES ACTUAL PROCESS MATCH WITH THE OPERATING INSTRUCTIONS?	N		
	7- DOES ACTUAL PROCESS MATCH WITH EXPECTED CORRECTIVE ACTION?	N		
	8- ARE OPERATORS AWARE OF THE DEFECT?	Y		
	9- ARE OPERATORS AWARE OF THE EXPECTED CORRECTIVE ACTION?	Y		
	10- HAS THE SOLUTION BEEN PROPERLY VALIDATED ON THE FLOOR?	N		
Visual Aid	11- IS THERE A VISUAL AID?	Y		
	12- IS IT POSTED ON A VISIBLE SPOT?	Y		
	13- IS THE CONTENT CLEAR?	Y		
Equipment	14- IS IT COHERENT WITH THE CONTENT OF THE CORRECTIVE ACTION?	Y		
	15- HAS IT BEEN MODIFIED? IS PREVENTIVE MAINTENANCE RECORD MODIFIED?	N		
	16- IS SET UP INSTRUCTION UPDATED? MATCHES WITH CONTROL PLAN?	N		
Quality Control Documentation	17- WAS THE PLAN CONTROL WITH THE CORRECTIVE ACTIONS UPDATED?	N		
	18.- WAS THE PROCESS FMEA UPDATED WITH THE CORRECTIVE ACTIONS?	N		
	19.- WAS THE WORK INSTRUCTION UPDATED WITH THE CORRECTIVE ACTIONS?	N		
	20.- WAS THE QUALITY REGISTER UPDATED WITH THE CORRECTIVE ACTIONS?	N		

AUDITOR'S NAME	Leticia Castillo	DATE OF THE AUDITS	27-09-2024					
Comments	Once the inductance is added to the tester, the internal documents, set up, Instructions, Master Parts	DECISION	Y N	N	Y N		Y N	

7.- Preventive Action

Preventive Action	Expected result efficiency	Who	Due Date	Done Date
Update the controls in the Self Control Sheet.	Strengthen the Quality system	Leticia Castillo	24-sep-24	27-sep-24
Update PFMEA	Strengthen the Quality system	Sughey Rangel	11-oct-24	11-nov-24
Update Control plan	Strengthen the Quality system	Sughey Rangel	11-oct-24	11-nov-24

8.- Standardize / Potential Applicability

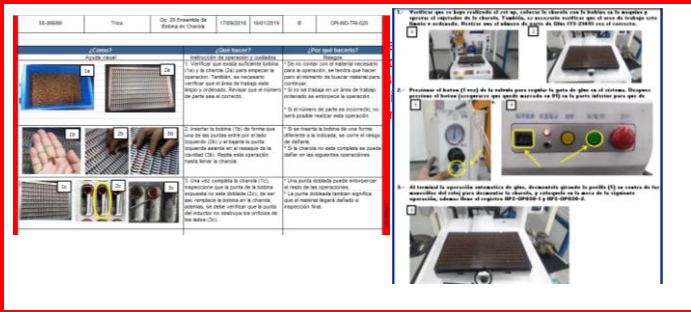
1.- Potential Applicability of Corrective / Preventive Action to similar lines / products.

To whom?			When?	
	Production Line:	All numbers parts	When?	11/10/2024
	Operation:	Parts molding	When?	11/10/2024
	Product:	All customers	When?	11/10/2024

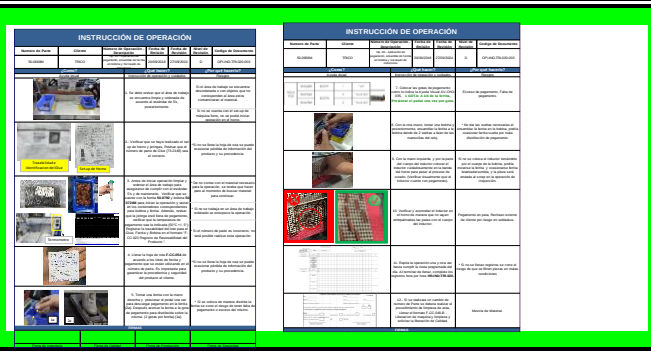
Comments

9.1.- Evidence your Efforts - Ocurrences

Before Corrective Action

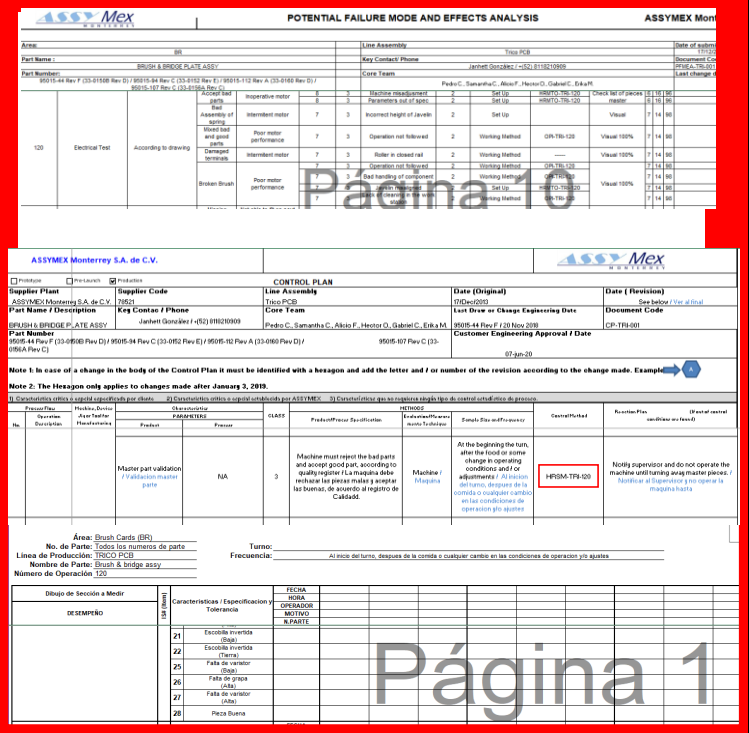


After Corrective Action

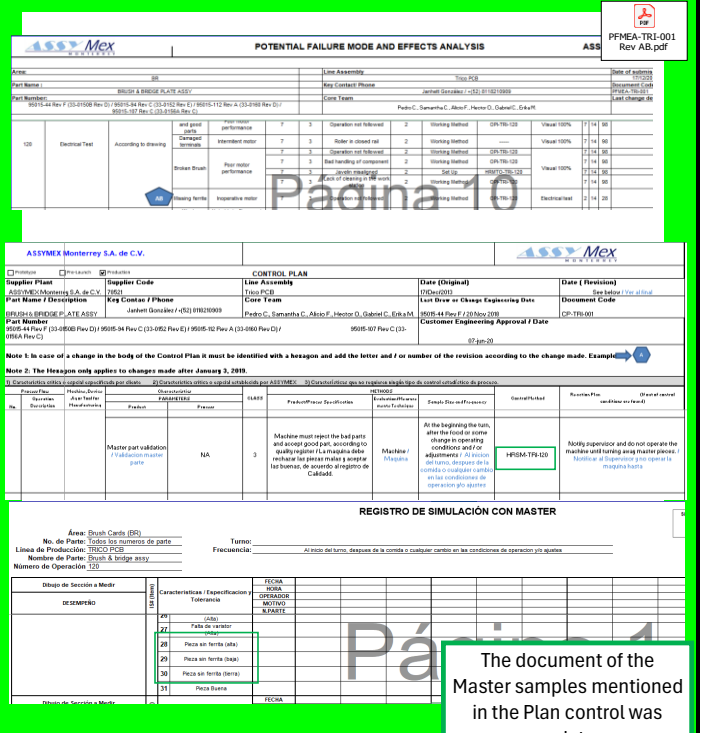


9.2.-	Evidence your Efforts - Non Detection

Before Corrective Action



After Corrective Action



10.-	Congratulate the Team